FINAL YEAR PROJECT | PROJECT REVIEW

SmartCricket AI

An Explainable Multimodal Cricket Shot Analysis and Intelligent Virtual Coaching System

BASE PAPER

CricShot10k: A Large-Scale Video Dataset for Cricket Shot Classification

M. K. Dihan, Abdullah, Amina and S. Ahmed | IEEE Access, vol. 14, 2026

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ACADEMIC YEAR
2026–2027

COLLEGE
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GUIDE
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Revised 5 November 2026 • feature expansion and evidence review

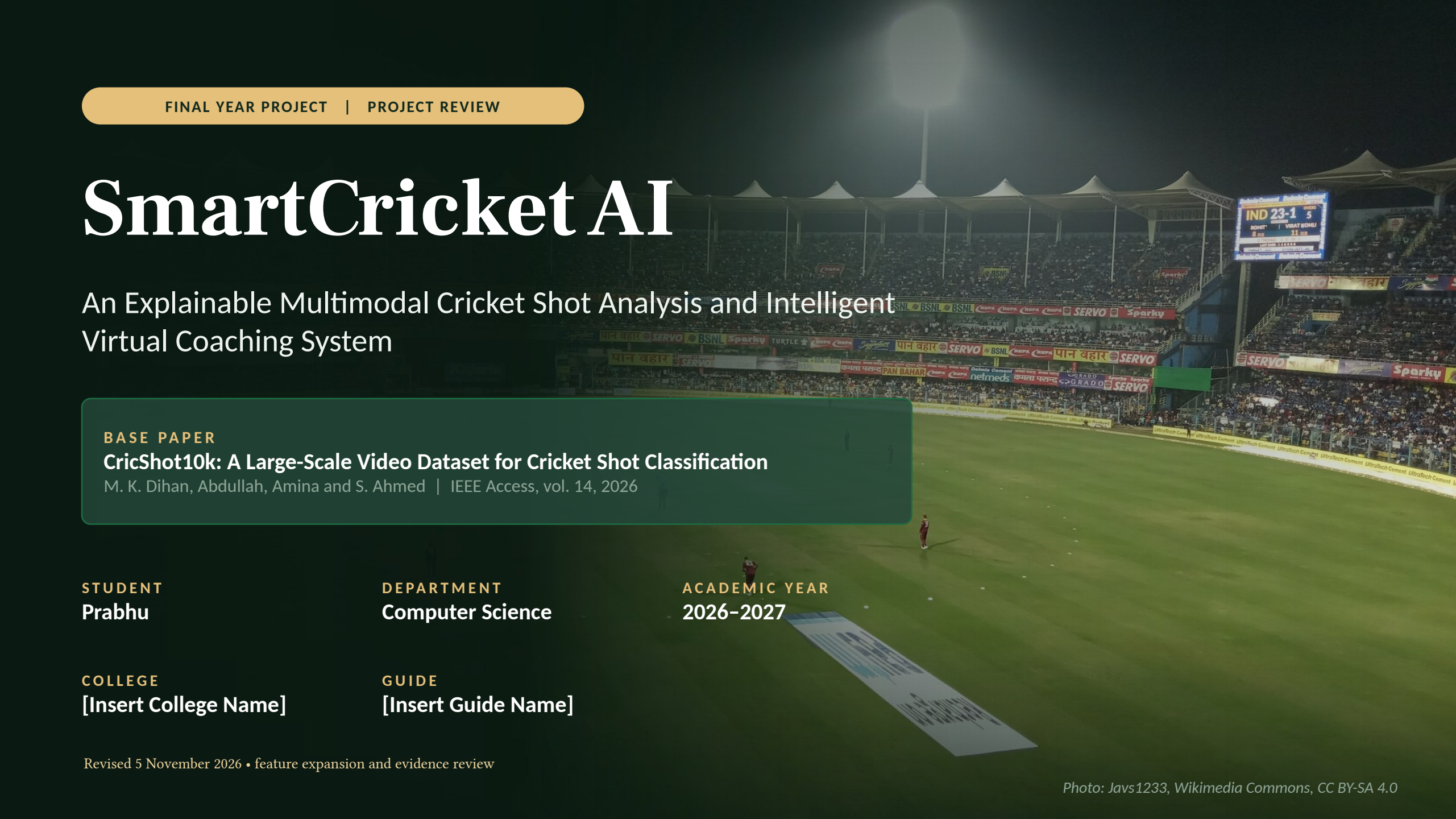


Photo: Javs1233, Wikimedia Commons, CC BY-SA 4.0

Abstract: The Project in One Slide

The problem

Batting shots are tagged by hand. It is slow, and players rarely get feedback on how they moved.

Starting point

The CricShot10k dataset: 10,086 one-second clips in 15 shot types, and its CricShotNet model.

What I built

A web app: photo posture report, video shot analysis, heatmaps, comparison with a reference clip, AI coach and daily training.

Main result

Saved local result: video 56.17%, fusion 63.93% on 1,289 test clips. Not rerun in this revision.



Photo: Acabashi, CC BY-SA 4.0; skeleton by SmartCricket AI

Why This Project



Photo: Acabashi, Wikimedia Commons, CC BY-SA 4.0

Manual work is slow

Someone has to watch every ball and type the shot. Across many matches and nets sessions this does not scale.

Feedback is rare

Club players and students seldom have a coach watching every session.

AI already helps

Tennis, football and baseball already use pose tracking and video models.

My idea

Build on published shot-recognition research and add explainable analysis and daily practice tools.

Project Objectives

1

Build a local baseline

Local EfficientNetV2-S + GRU baseline; not a full reproduction of the paper.

DONE

2

Add body pose

Find 33 body points with MediaPipe and turn them into joint angles.

DONE

3

Compare the models

Video only, pose only and video + pose on the same test clips.

DONE

4

Build photo and video modes

One web app: HTML, CSS, JavaScript front end and FastAPI back end.

DONE

5

Explain and coach

Heatmaps, comparison with a reference clip, an AI coach and a daily training plan.

DONE

Base paper and project contribution

Paper: dataset and classifier

10,086 clips • 15 classes • 89% reported Top-1. DOI: 10.1109/ACCESS.2026.3663220

Local recorded experiment

Video 56.17% → video + 2D pose 63.93% Top-1. Different experimental setup.

Application contribution

Photos, explanations, reference comparison, daily practice, goals, retests and reports.

New in this revision

Five synthetic demo scenarios, 70 fictional practice days, labeled exports and validation-only model selection.

A broader feature set is a project contribution. Higher accuracy than the base paper has not been demonstrated.

Existing System: CricShotNet (Base Paper)

Broadcast clip
1 second

→

Crop
YOLOv11 finds batter

→

Sample
15 frames

→

Tint
bat and batter

→

EfficientNetV2-S
reads each frame

→

GRU (128)
follows the swing

→

Answer
1 of 15 shots



Crop and tint: Figs. 11-12 of the base paper, CC BY 4.0

What the authors report

89%
top-1 accuracy

97%
top-2

99%
top-3

Cropping + tinting: 89.1% vs 85.8% with cropping only.

Hardest pair: pull and hook (13% confused).

Existing System: Disadvantages

No explanation

It gives a label and a number, but not why.

No app

The paper focuses on research data and classifier code, rather than this player workflow.

Broadcast video only

Not tested on phone or nets videos.

Pose not used

No explicit pose measurements or joint-angle posture report.

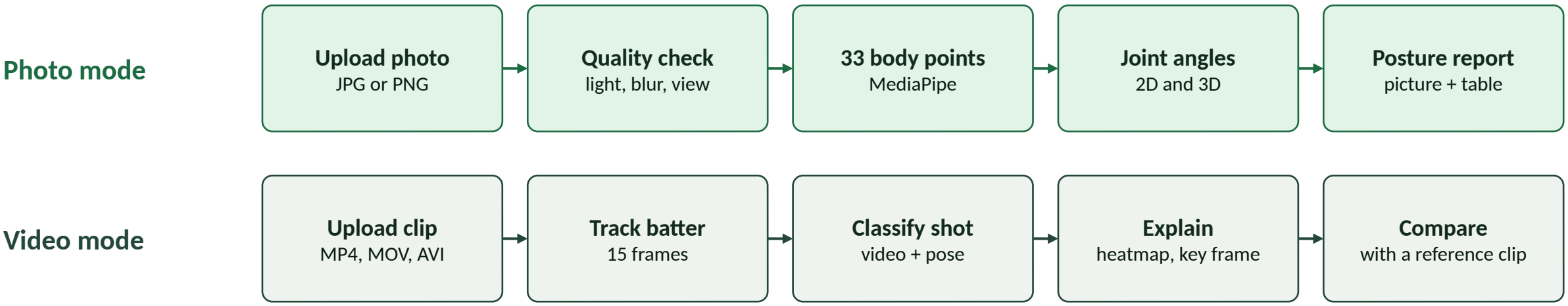
No photo support

Many players only have a photo of their stance.

Confidence not checked

A score of 90% may not really mean 90% right.

Proposed System: Two Ways to Analyse a Batter



AI Coach

Uses stored facts and numeric consistency checks, with optional voice playback.

Daily training

Daily plans from stored measurements and check-ins, with recovery, activity levels and reports.

Live mirror

Camera skeleton and rep counts; optional recognition or saving can upload frames.

All Features at a Glance

Photo posture report

Angles, visibility and a 3D view from one photo.

Video shot analysis

Shot name, calibrated scores and top-3 candidates.

Heatmaps

Shows where on the frame the model looked.

Key-frame timeline

Shows temporal attention across sampled frames.

Reference comparison

DTW differences against an admin-provided reference clip.

Movement analysis

Movement analysis is available without a classifier.

AI Coach + voice

Uses stored facts with numeric consistency checks.

Training coach

Daily drills, check-in, quiz, levels and XP.

Live mirror

Live skeleton, rep count, Shadow the Pro.

Phone mode

Scan a QR code and use the phone camera.

History and PDF

Every result saved; one-click PDF report.

Admin console

Players, reference clips and model results.

Proposed System: Advantages

Photo or video

A quick check from a photo, a full analysis from a clip.

Inspect model behavior

Available heatmaps, attention and skeleton views help inspect results.

Flags low confidence

Scores below the threshold receive an uncertainty warning.

Plain-language feedback

The coach separates 'Measured' facts from 'General tip'.

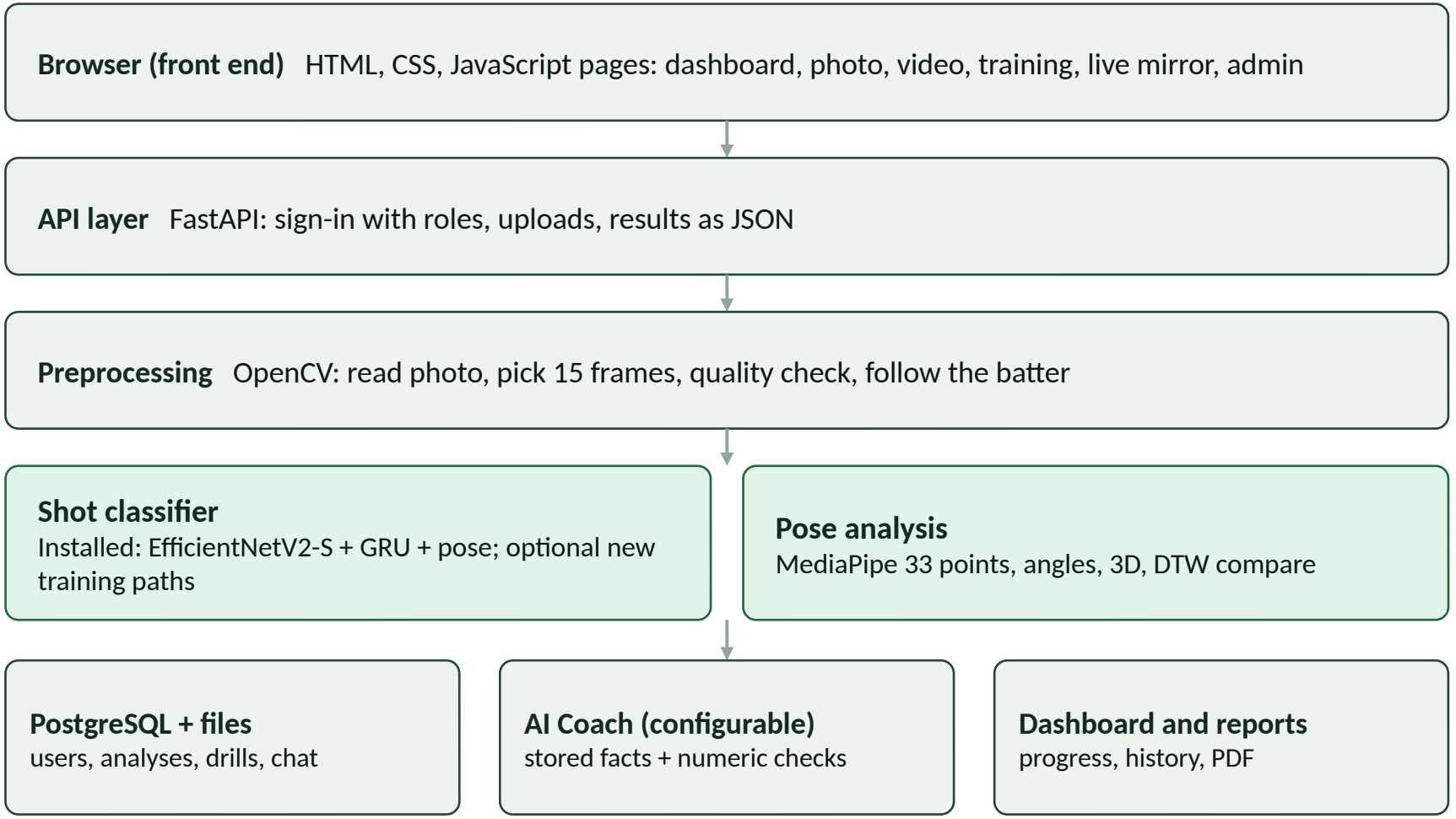
Private

Pose runs locally; optional recognition and saved moments upload frames.

Low cost

Open-source stack and Colab training workflow; actual compute cost varies.

System Architecture



Software Description

Front end

HTML5, CSS3

Vanilla JavaScript

Chart.js charts

Web Speech (voice)

Back end

Python FastAPI

Uvicorn server

PostgreSQL

JWT sign-in, roles

Deep learning

PyTorch

EfficientNetV2-S

GRU + attention

Temperature scaling

Vision

OpenCV

MediaPipe Pose (33 points)

Grad-CAM heatmaps

DTW comparison

Tools

Google Colab (T4 GPU)

VS Code, Git

pytest + Node checks

Gemini API (coach)

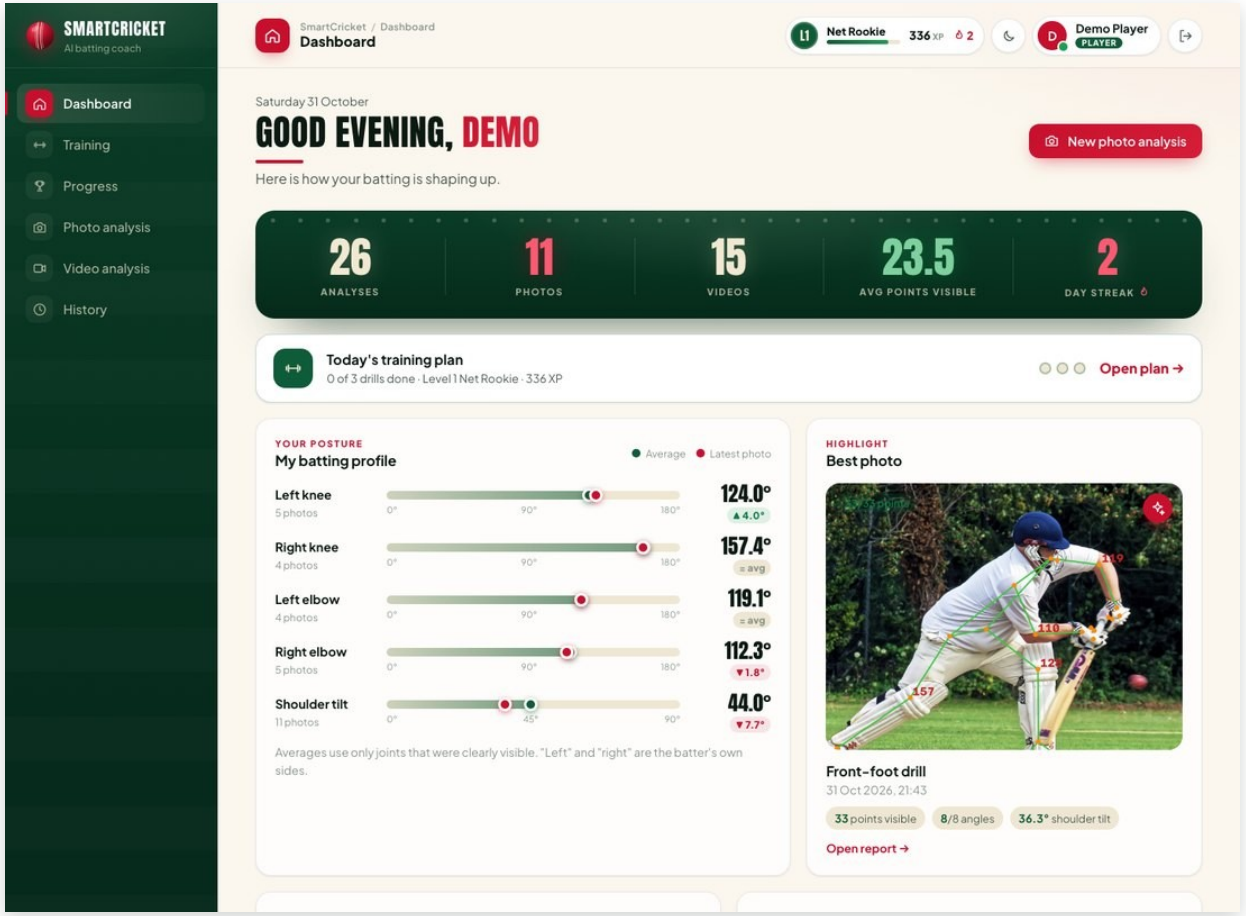
M1 User Interface and Sign-in

Two roles: player and admin, each with their own pages.

Secure sign-in: hashed passwords and tokens; sign-out clears the session.

Dashboard: next steps, today's plan, saved measurements and clearest photo.

Works on phones: the layout adapts to any screen size.



Earlier app screenshot; isolated/example data. See revised feature slides.

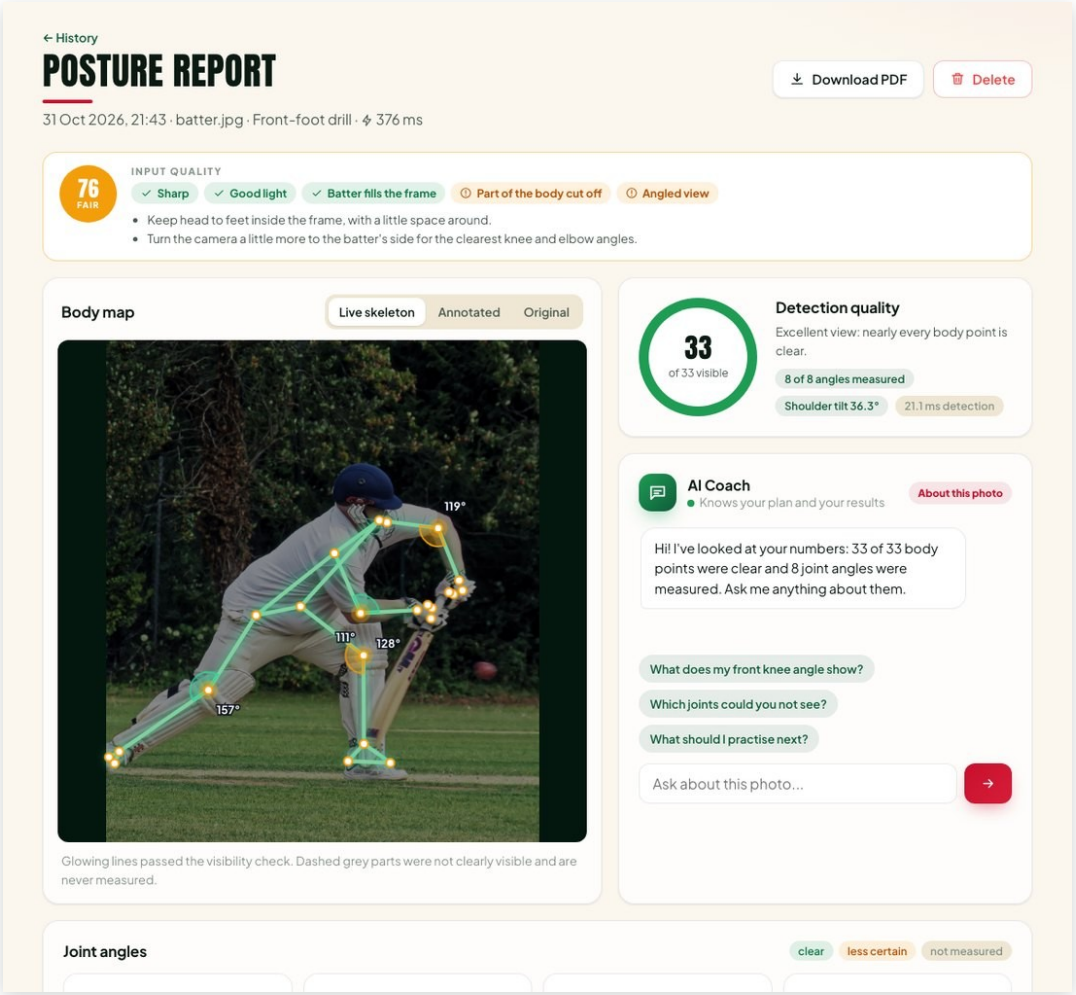
M2 Photo Posture Report

Quality check first: light, blur, body size and camera angle, with a tip to fix each.

33 body points: estimated by MediaPipe; below-threshold joint measurements are omitted.

Joint angles: knee, elbow, shoulder and hip, in 2D and 3D.

No shot guess: one photo cannot show the swing, so it reports posture only.



Earlier app screenshot; isolated/example data. See revised feature slides.

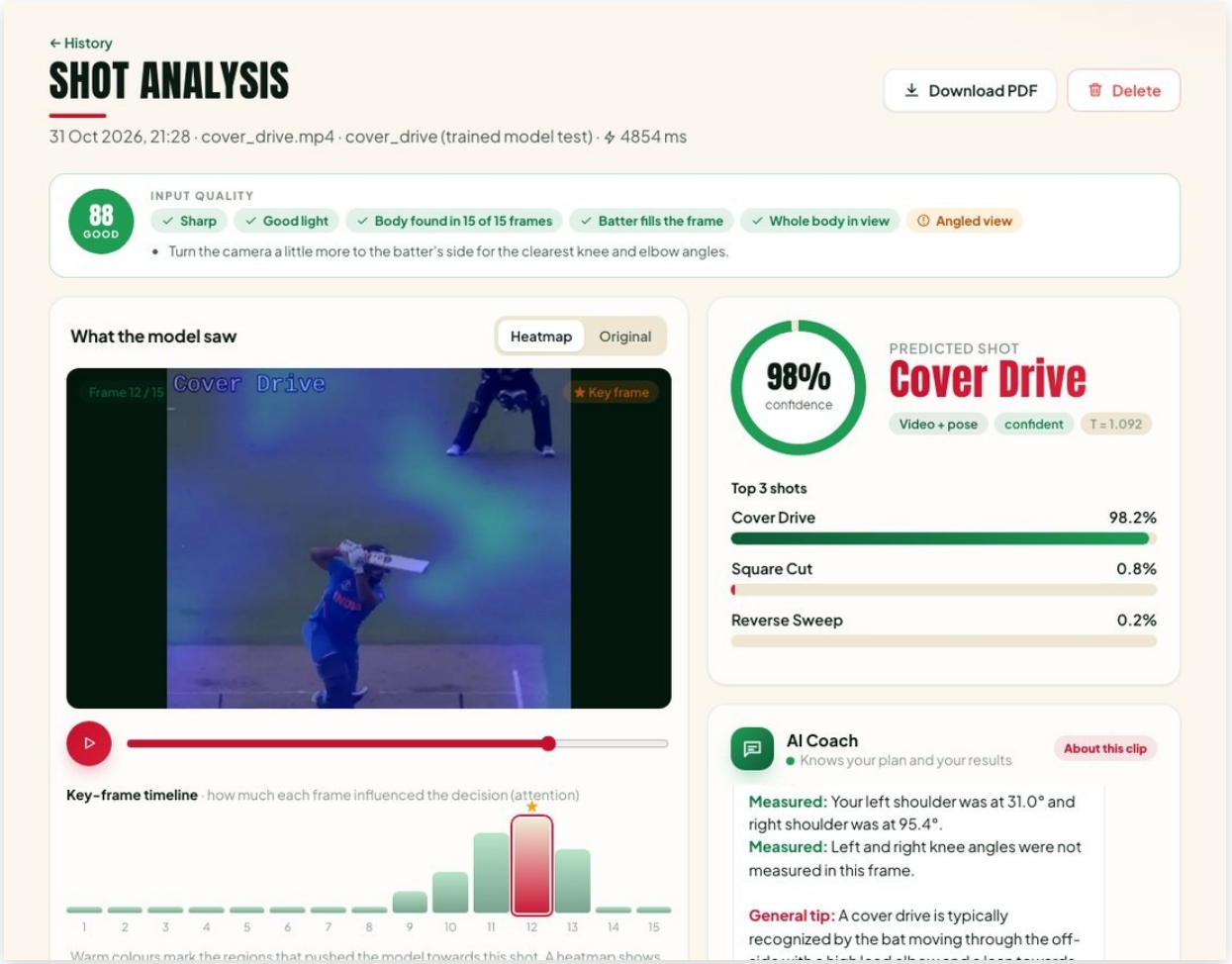
M3 and M4 Video Shot Analysis

Prepare: check the file, follow the batter, pick 15 frames.

Classify: video + pose model names 1 of 15 shots with top-3 answers.

Calibrated scores: temperature-scaled confidence; individual predictions can still be wrong.

Explain: heatmap on the frame and a key-frame timeline.



Earlier app screenshot; isolated/example data. See revised feature slides.

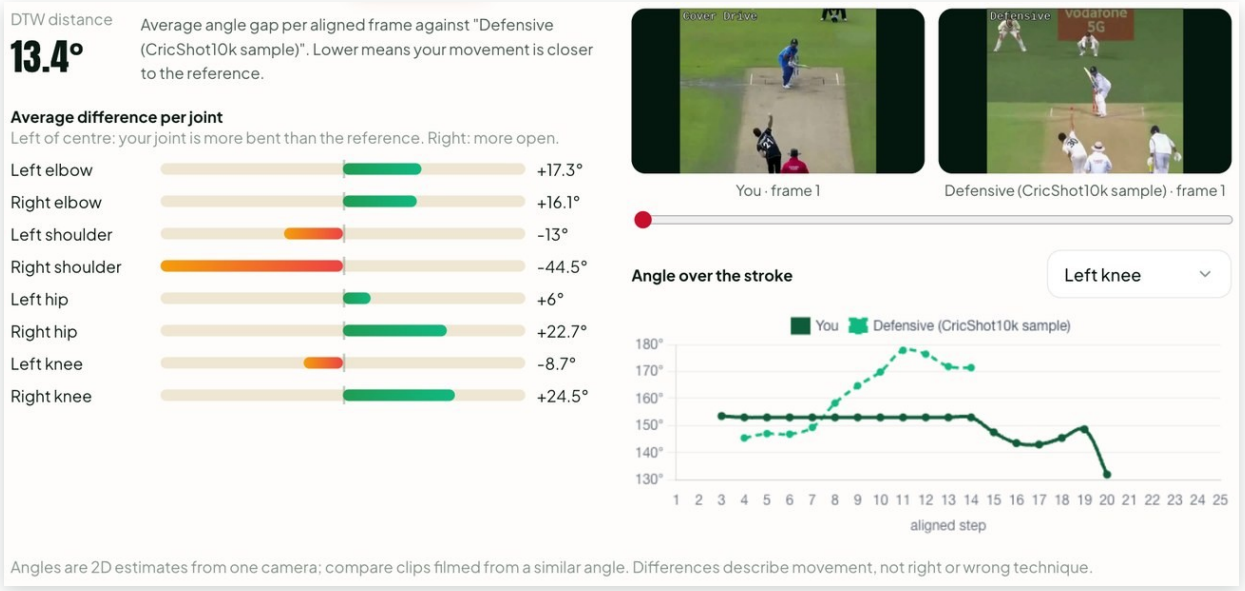
M5 Compare with a Reference Clip

Pick a reference: an administrator-provided example of the same shot.

DTW alignment: lines up both swings even if one is faster.

Joint-by-joint gap: shows which joint differs most, in degrees.

3D view: rotate the body skeleton to see it from any side.



Earlier app screenshot; isolated/example data. See revised feature slides.

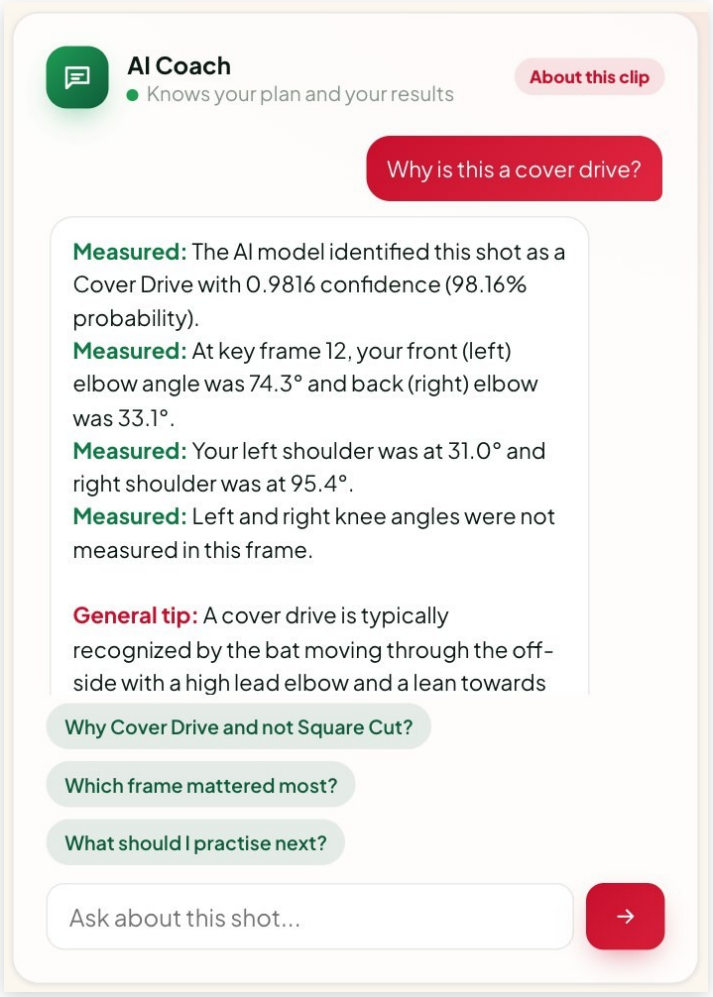
M6 and M7 AI Coach

One coach everywhere: on results, training and progress pages.

Grounded: stored facts and a number guard reduce unsupported values; no correctness guarantee.

Two kinds of line: 'Measured' facts and 'General tip' advice.

Voice: reads answers aloud and takes spoken questions.



Earlier app screenshot; isolated/example data. See revised feature slides.

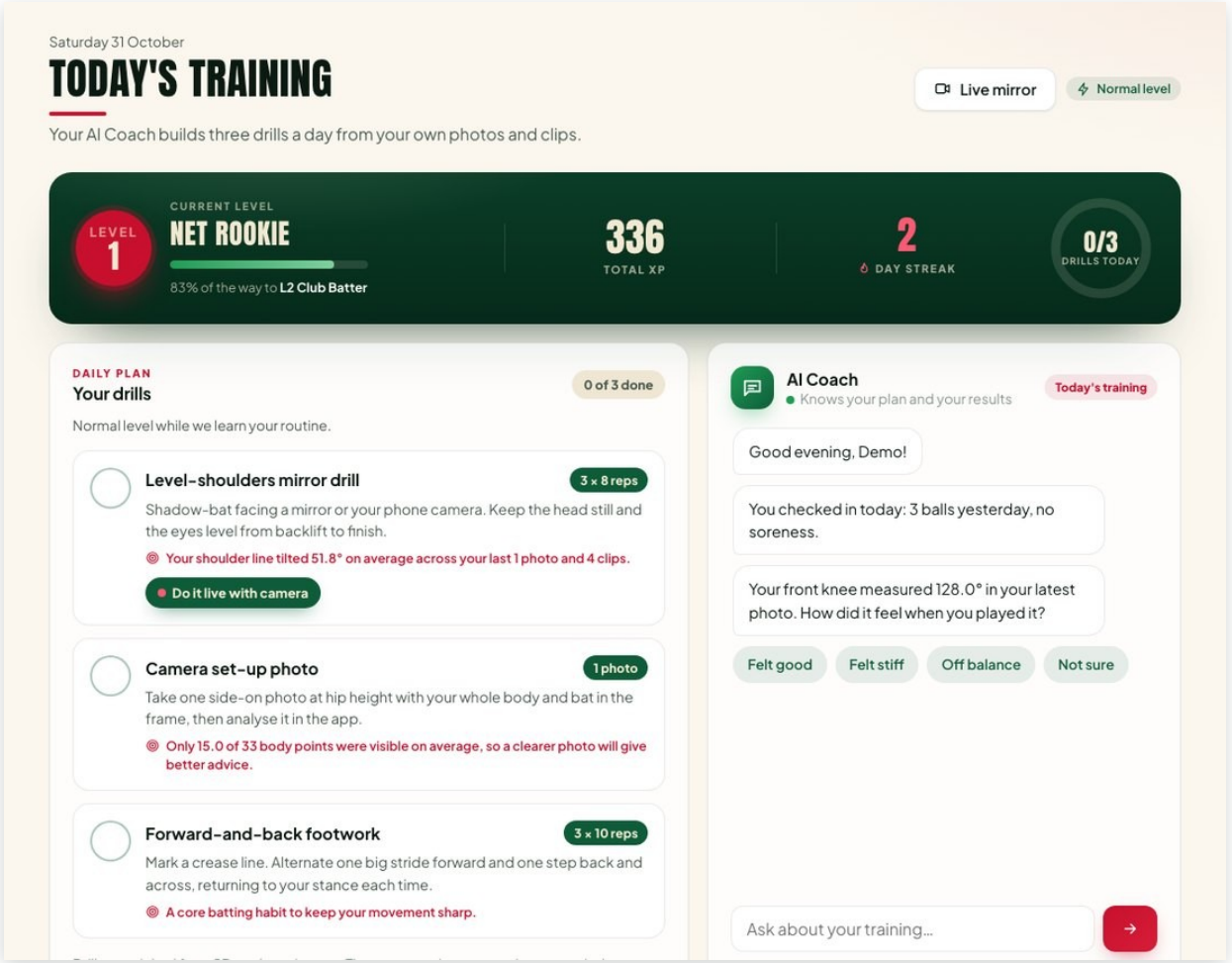
M9 Daily Training Coach

Three drills a day: chosen using recent measurements and the current check-in.

Adapts: check-in about feeling and soreness; recovery overrides where needed.

Motivation: activity levels, XP, streaks and a daily quiz; not verified batting skill.

Weekly report: summary written by the coach every week.



Earlier app screenshot; isolated/example data. See revised feature slides.

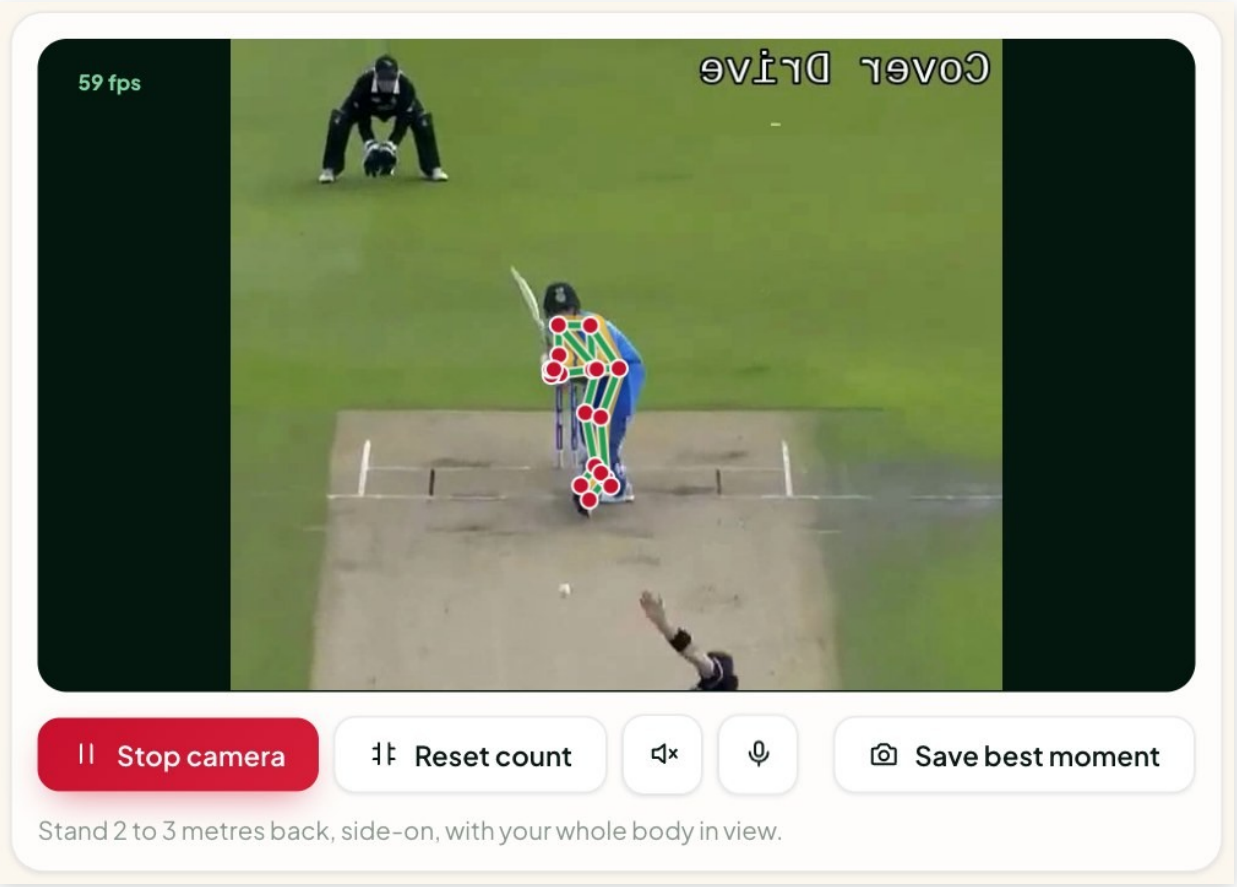
Live Mirror and Phone Mode

Live skeleton: browser pose tracking; optional recognition sends frames and saving uploads one.

Counts reps: and speaks corrections; voice commands start and stop it.

Shadow the Pro: an overlay from an administrator-provided reference.

Phone mode: scan a QR code and use the phone camera in the nets.



Earlier app screenshot; isolated/example data. See revised feature slides.

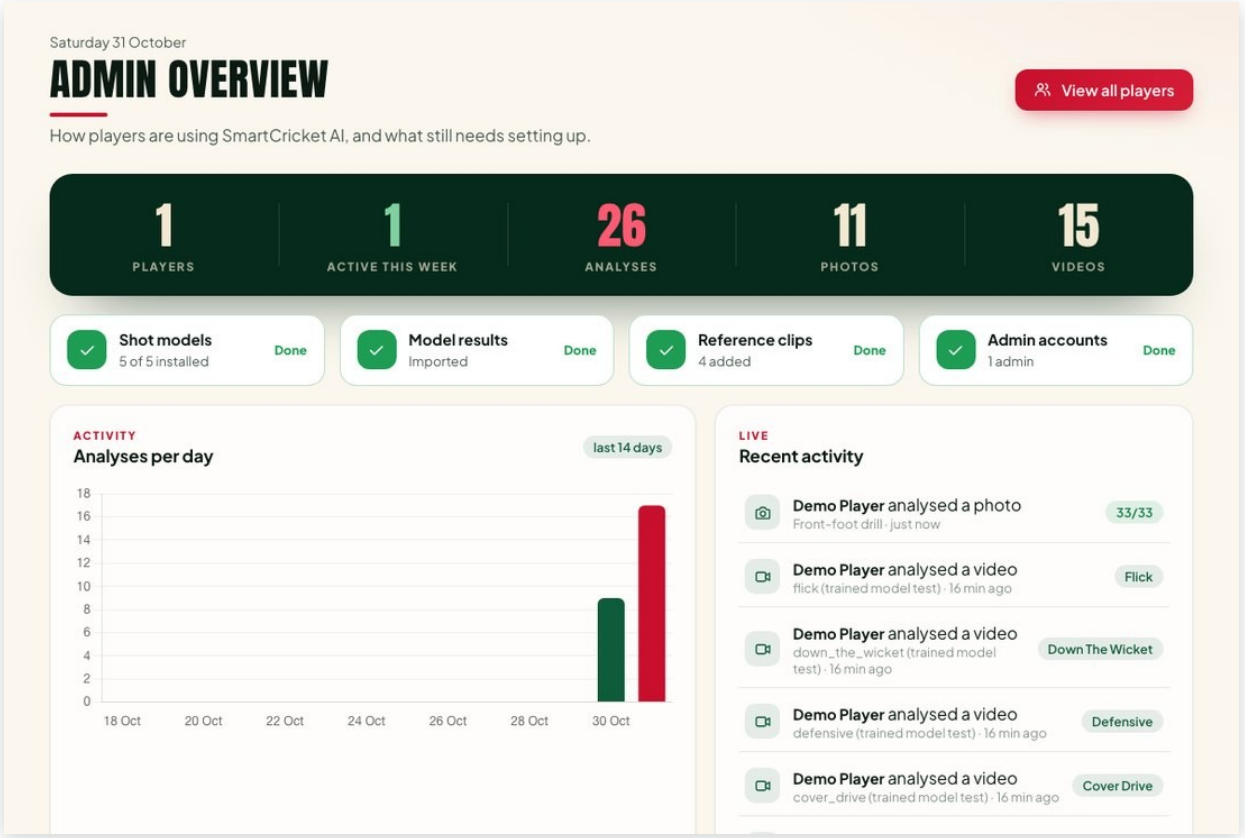
M8 History, Reports and Admin

History: every photo and clip saved with its result; search and filter.

Reports: one-click PDF and CSV export.

Progress: trends, weekly numbers and levels.

Admin console: players, reference clips, setup checklist and model results.



Earlier app screenshot; isolated/example data. See revised feature slides.

Recorded results: does pose help locally?

56.17%

Local video Top-1

63.93%

Local fusion Top-1

+7.76

Percentage points

1,289 recorded test clips • one stored seed per model

Fusion Top-3: 89.60% (a different metric from Top-1)

Paper: 89% Top-1 under its own setup; no superiority claim

Source: data/results/summary.json. Training and test inference were not reproduced for this revision.

Results: Trust, Robustness and Speed

Honest confidence (E5)

Recorded ECE decreased for all models. This does not guarantee any individual prediction.

Camera angle (E10)

Pose helped most for front-on clips: 47.3% to 57.6% macro-F1.

Pose reliability (E6)

Shoulders and hips found in about 69% of frames; wrists are hardest.

Speed on a laptop CPU

Historical timings depend on device and mode; this revision did not rebenchmark speed.

Heatmaps (E7)

Clear focus on bat and batter for some shots; weak for others, reported honestly.

Testing

See the revision verification record for current Python, Node and browser checks.

Practice goals: baseline → retest

Save a baseline and personal focus

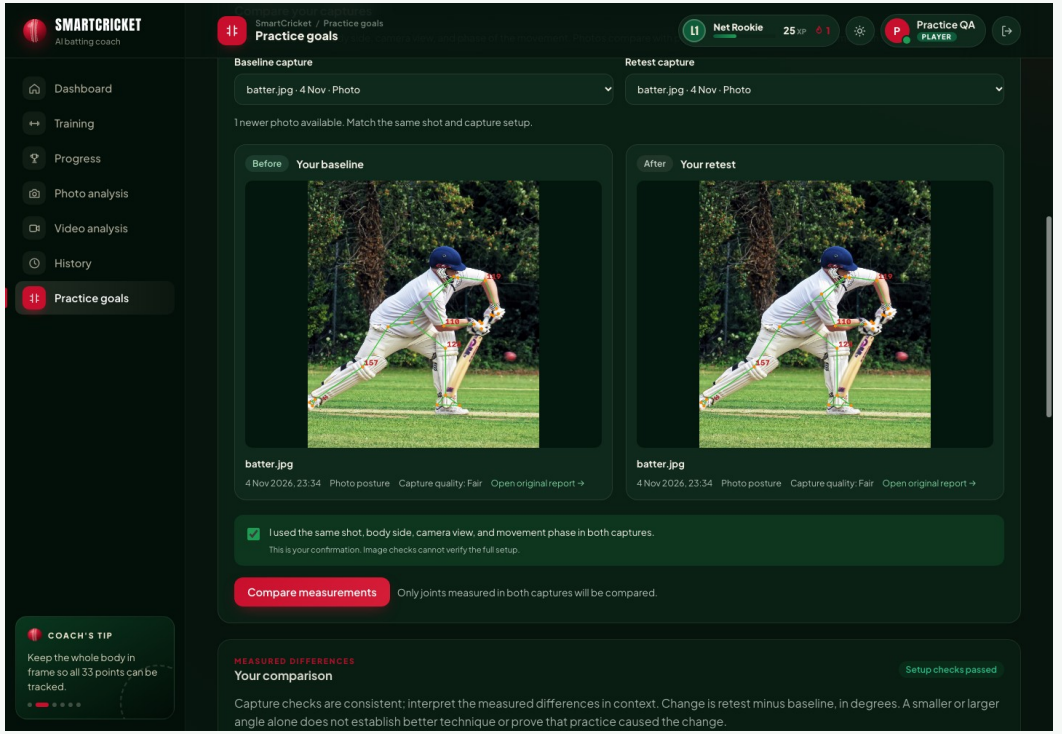
Practise with goal context

Capture a newer photo or clip

Select comparable video frames

Review deltas, visibility and setup checks

Save the retest and download the PDF



Angle change and target distance are measurements. They do not establish better technique or a causal practice benefit.

SYNTHETIC EXAMPLES ONLY

New: a clearly labeled demo showcase

Five selectable scenarios

Closer target, farther target, camera mismatch, limited visibility and recovery.

70 fictional practice days

Drill planning and completion, with rest days handled separately.

Reviewable arithmetic

Signed joint deltas, shared visibility, personal target distances and warnings.

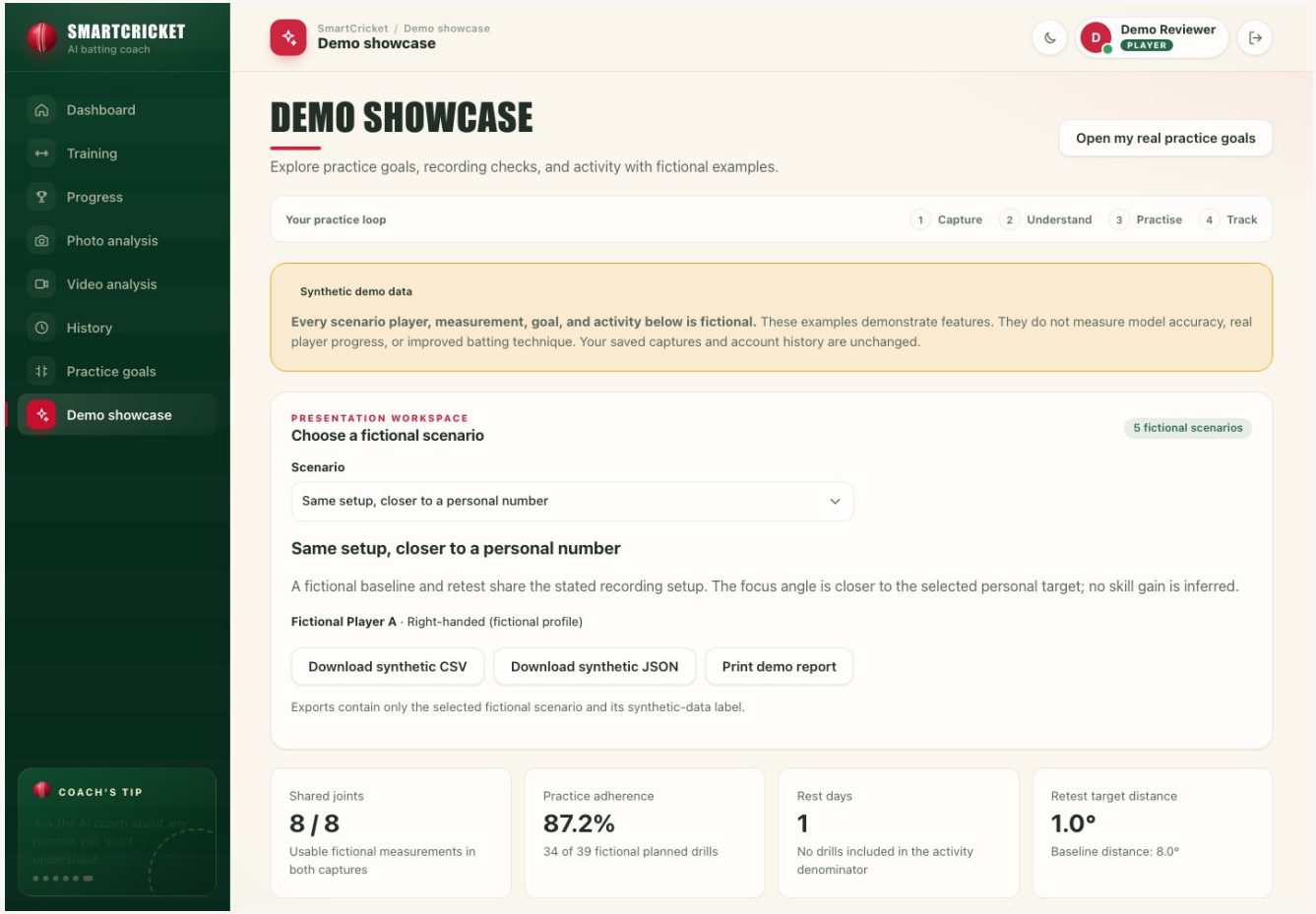
Portable disclosure

CSV and JSON retain synthetic provenance. No model accuracy is generated.

Player menu → Demo showcase

All data is bundled; no player histories or saved evaluations are changed.

Demo showcase in the application



- Persistent synthetic-data notice
- Five scenario states
- Personal-target distances
- CSV / JSON / print exports
- No research accuracy generated

How to read a demo comparison

Left knee: 148° → 141° | Retest − baseline = −7°

Personal target: 140°

Distance from target: $|148 - 140| = 8° \longrightarrow |141 - 140| = 1°$

The constructed retest is closer to a chosen target.

A smaller target distance does not establish improved batting skill.

Camera mismatch or low visibility still limits comparison.

Every value on this slide is synthetic. It is excluded from classifier evaluation.

Evidence limits to resolve

Dataset count

Local split total: 10,091. Paper: 10,086. Source manifests are needed to reconcile this.

Statistical ordering

Saved paired-test labels conflict with the accuracy direction. Recompute from paired predictions.

One recorded seed

Saved metrics are single-run artifacts. No new training or test inference was done here.

Calibration and coaching

Lower ECE and numeric guards help; neither guarantees correctness on a new input.

Keep recorded research results, functional checks and synthetic demonstrations separate in the review.

A valid path to higher real accuracy

1. Reconcile genuine clips and labels; remove source/near-duplicate leakage.
2. Freeze train, validation and test manifests. Augment training only.
3. Compare cropping, stronger features and fine-tuning against the baseline.
4. Select model and seed on validation before held-out testing.
5. Save per-clip predictions; report repeat seeds, calibration and per-class errors.
6. Update the PPT and report only after new measured results exist.

Conclusion and Future Work

What was achieved

Built: a complete web app with photo, video, live and coaching features.

Recorded: saved local fusion accuracy exceeds the local baseline by 7.76 points.

Trustworthy: calibration and explanation aids, plus a coach with numeric consistency checks.

Useful: daily training plan, progress tracking and PDF reports.

Future work

- Reconcile data provenance; evaluate cropping and fine-tuning on fixed splits
- Video transformers (Video Swin, TimeSformer)
- More shot classes only after collecting genuine labeled clips
- A mobile app and live match streams

References

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[8] C. G. Rao et al., "Cricket shot classification from video using body pose landmarks," Proc. ICAECT, 2025.

[9] A. Sen et al., "CricShotClassify," Sensors, vol. 21, no. 8, 2021.

[10] S. W. Hoque et al., "KUCricShot: A dataset for developing detection model of cricket shots," Proc. ICCIT, 2023.

Thank You

Questions and live demo

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